AMPERES

Haunted Chess

A robot arm that sees your move, answers with Stockfish, taunts you, sets the board up itself, and audits every move as training data.



github.com/vnmoorthy/amperes-haunted-chess

Robot data fails silently

Hours of episodes

Robot-learning teams pay for every station hour. Some episodes come out unusable: missed grips, wrong labels, camera gaps. Nobody notices until training.

Chess is the test bench

Known object, known target, an independent label from the game. Six piece shapes, crowded squares, captures, castling: real manipulation, judged by a rulebook.

So we built both

A robot that plays real chess, and an audit layer that turns every pick-and-place into an episode with a verdict and the evidence behind it.

What you see in the live demo

1

Robot opens

Stockfish picks; the arm plays, fast in the air, gentle near pieces

2

You reply by hand

No typing: the wrist camera finds the legal move you made

3

It answers and taunts

Claude writes a spooky line; the voice reads it as the arm moves

4

Every move is data

Episodes with verdicts in the Amperes console, LeRobot export

“C four! I offer a pawn, like candy from a haunted house.”

How it sees your move



- Locks the printed grid from 56 cm: all 64 squares, 0.5 px fit
- Depth says which squares hold a piece; brightness says whose
- Finds the one legal move that explains the change, seen twice
- Castling, captures and en passant matched the same way

It sets up the board by itself

Six-view survey

The wrist camera visits six poses over the board and 10 cm around it; every piece becomes a 3D blob in arm coordinates.

Named twice

Claude names each zoomed piece; its measured height must agree. A king is 85 mm, a pawn 45.

Clearance planner

Short pieces first; a tall neighbour is lifted aside and put back, like a person does.

600/600

simulated scrambles of 34 pieces reset with zero gripper contact

14/14

real pick-and-places in a row after calibration

4.5 mm

survey position error, down from 15 mm

It catches its own mistakes

Empty grip

Finger gap reads -2 on air vs 12+ mm on a piece: retry at once, nudged and lower

Phantom move

If the camera does not see the move, it is redone, never recorded as played

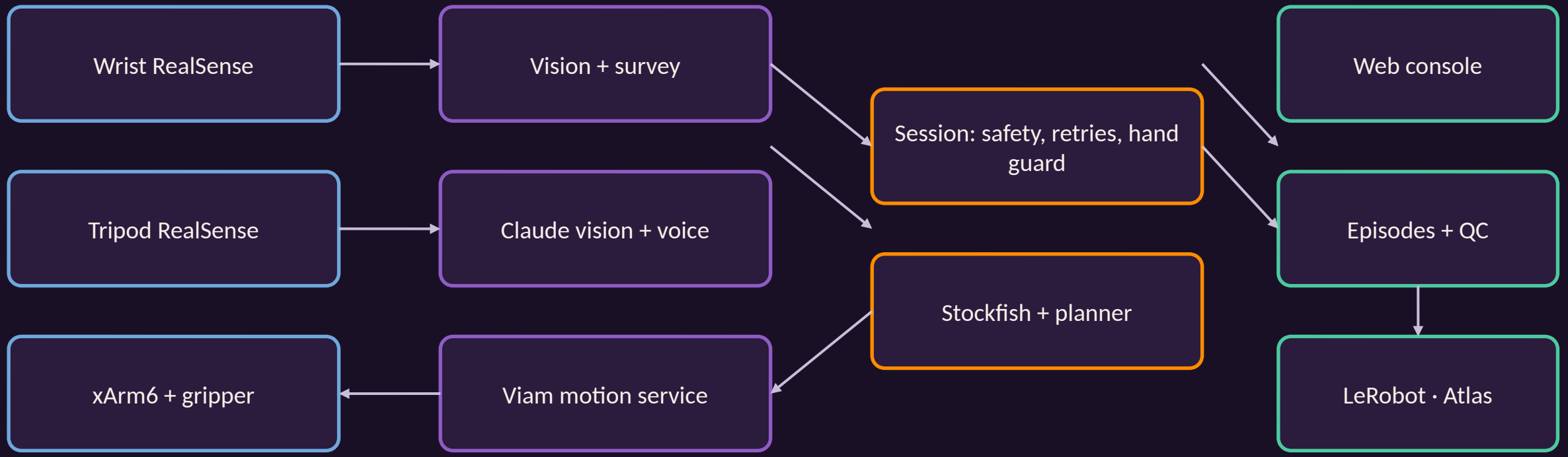
Camera bias

Measured on pieces the arm placed itself: a 15 mm offset, corrected by a fitted transform

Hands

Before every descent the camera checks under the gripper and holds still for anything tall

Architecture



All Python, one repo, 130 tests. Every module is reusable on any Viam arm.

Every move becomes audited data

Episode + verdict + evidence

Each pick-and-place is an episode: usable, suspicious, invalid or failed-but-valid, with the evidence. Yesterday it caught the gripper reporting “got it” while the camera showed the pawn never moved.

Ready for robot learning

Exported to LeRobot format, synced to MongoDB Atlas, searchable across stations. Third-party LeRobot datasets from Hugging Face go through the same audit.

Measured on the event arm

10/10

robot moves confirmed by camera in game 1

0.34 mm

max error over 186 commanded poses

130

automated tests passing

~3 mm

board calibration offset, measured and corrected

6 views

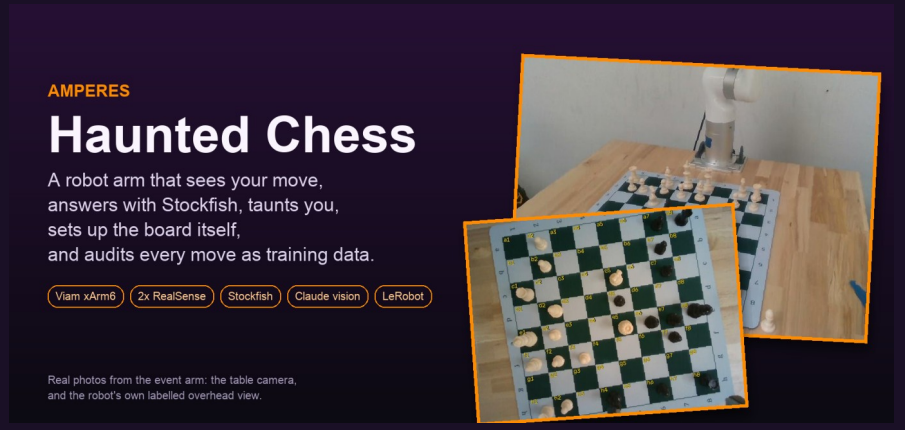
board survey in under a minute

0 contact

in 600 simulated board resets

Built to be reused

- chess_lifecycle: the clearance-aware planner, pure Python
- chess_survey: multi-view scan with a fitted camera correction
- chess_vision: grid lock and human-move detection
- Amperes audit: QC and LeRobot export for any robot



Play it. Fork it. Haunt it. 🎃

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